# VHF / UHF Handheld — Channel & Net Reference

RX 108–136 (AM air) · 136–174 · 350–390 · 400–520 MHz  TX 136–174 · 400–480 MHz (FM, 5 W)

Indianapolis Metro

GMRS ch → GMRS licence · Ham 2m/70cm → Technician · MURS/FRS licence-free



## WEEKLY NET SCHEDULE (times Eastern; * = unverified, confirm on air)

| Day | Time | Net | Ch# | Freq | Tone / Access |
|---|---|---|---|---|---|
| Sun | 7:00p | Johnson ARES/Mid-State ARC | 335 | 146.835 | PL151.4 |
| Sun | 7:00p | Michigantown 625 [GMRS] | 126 | 462.625 | PL131.8 |
| Sun | 7:30p | Marion Co ARES | 343 | 146.700 | PL88.5 |
| Sun | 8:00p | Hamilton Co ARES / CIARA | 350 | 145.170 | PL77.0 |
| Sun | 9:00p | Anderson WA9EOC net * | 615 | 444.675 | CSQ |
| Mon | 7:00p | Central Indiana GMRS Net [GMRS] | 120 | 462.575 | **Courtesy req.** |
| Mon | 7:15p | Marion Co CERT Net | 623 | 443.250 | PL100.0 |
| Mon | 7:30p | Blue River Valley ARC/ARES | 358 | 145.480 | PL88.5 |
| Mon | 7:30p | Emergency Simplex Test ch19 [GMRS] | 128 | 462.650 | simplex |
| Mon | 7:30p | Fusion Tech MinWis [DIGITAL] | 620 | WIRES-X | C4FM |
| Mon | 8:00p | Tipton County Net (K9TRC) | 637 | 443.125 | PL131.8 |
| Mon | 8:00p | Crossroads DMR [DIGITAL] | - | TG 31189 | DMR |
| Mon | 9:00p | Mesh Monday (DRP) [ZELLO] | - | Zello app | mesh net |
| Mon | 11:00p | The Papa Chat [DIGITAL] | - | TG 31077 | DMR |
| Tue | 7:30p | Hancock County (W9ATG) | 617 | 444.450 | DCS465 |
| Tue | 7:30p | Hendricks County ARES | 334 | 145.130 | PL88.5 |
| Tue | 8:00p | Absolute Tech Net (natl) * | 356 | 146.745 | D346 |
| Tue | 8:00p | Mid-State WIRES-X [DIGITAL] | 619 | 443.525 | C4FM |
| Tue | 8:00p | Hoosier DMR Net [DIGITAL] | - | 441.200 | TG 3118 |
| Tue | 9:00p | INDY 600 net [GMRS] | 122 | 462.600 | **$25/yr if regular** |
| Wed | 7:00p | Newsline bulletin (W9IRA) | 343 | 146.700 | PL88.5 |
| Wed | 7:30p | Marion County ARES | 343 | 146.700 | PL88.5 |
| Wed | 8:00p | Welcome Net (W9ICE) | 346 | 146.970 | PL107.2 |
| Wed | 8:00p | Madison Co ARC (W9VCF) | 329 | 145.390 | PL151.4 |
| Thu | 6:30p | EchoLink Youth Net | 356 | 146.745 | D346 |
| Thu | 7:00p | SATERN (Salvation Army) | 345 | 146.880 | PL88.5 |
| Thu | 8:00p | Indiana Statewide / AUXCOMM | 339 | 147.210 | PL88.5 |
| Thu | 8:00p | IN Statewide D-STAR [DIGITAL] | - | REF024 B | D-STAR |
| Thu | 8:00p | Who Knows What Net (K9EOC) | 620 | 443.950 | PL77.0 |
| Sat | 8:00p | Doomsday Preppers Net [ZELLO] | - | Zello app | internet PTT |

## MONTHLY · HF / STATEWIDE ([HF] = beyond this handheld — needs HF rig or SDR)

| | | | | | |
|---|---|---|---|---|---|
| 1st Fri | 10:50a | Marion Co Siren Check | 343 | 146.700 | PL88.5 |
| 1st/3rd Sun | 8:00p | WC9IN Sunday Night Net (linked) * | 356 | 146.745 | D346 |
| 1st/3rd Thu | 7:30p | AUXCOMM HF pre-net [HF] | - | 7.250 / 3.844 | SSB |
| Daily | — | Indiana Traffic Net [HF] | - | 3.910 | SSB |
| Daily | — | Indiana Section CW Net (QIN) [HF] | - | 3.535 | CW |
| Sun | 5:00p | IN Section ARES HF Net [HF] | - | 3.900 / 7.280 | SSB |
| Sat | 9:00p | WVARA HF Ragchew [HF] | - | 3.679 | LSB |

## ANTENNA CUT LENGTHS (in.)

| Band | MHz | 1/4-wave | 1/2-wave |
|---|---|---|---|
| 10 m ham | 28.4 | 98.9" | 197.7" |
| 6 m ham | 52.5 | 53.5" | 107.0" |
| Airband RX | 121.5 | 23.1" | 46.2" |
| 2 m ham | 146.0 | 19.2" | 38.5" |
| MURS | 151.9 | 18.5" | 37.0" |
| Marine/NOAA | 159.5 | 17.6" | 35.2" |
| 1.25 m ham | 223.5 | 12.6" | 25.1" |
| 70 cm ham | 440.0 | 6.4" | 12.8" |
| GMRS / FRS | 462.6 | 6.1" | 12.1" |
| LoRa mesh | 915.0 | 3.1" | 6.1" |

*1/4-wave in. = 2808 / MHz (0.95 vel. factor)*

## RADIO SETTINGS (known-good)

| | |
|---|---|
| Squelch | 2 |
| Timeout timer | 120 s |
| Roger beep | **OFF** |
| DTMF / ANI ID | **OFF** |
| Key beep | OFF |
| Side tone | OFF |
| VOX | OFF |
| Tail Clear (STE) | ON |
| Scan mode | Carrier |
| Work mode | Channel |
| Save mode | ON |
| Dual watch | your pref |

*Roger + ANI OFF — required on some repeaters.*

## POWER · OFFSETS · CALLING

**GMRS TX POWER LIMITS**

ch 1–7 462.5625–462.7125 5 W
ch 8–14 467 MHz (FRS only) 0.5 W
ch 15–22 462.550–462.725 50 W
MURS 151 / 154 MHz 2 W

**REPEATER OFFSETS**

2 m ± 0.600 70 cm + 5.000
GMRS ch15–22 + 5.000 (in 467.x)

**CALLING / EMERGENCY**

2 m simplex call 146.520
70 cm simplex call 446.000
GMRS travel/emerg. 462.675 PL141.3
Marine distress ch16 156.800
Aircraft emergency 121.500

## NATO PHONETIC ALPHABET — for net check-ins

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **A** Alfa | **B** Bravo | **C** Charlie | **D** Delta | **E** Echo | **F** Foxtrot | **G** Golf | **H** Hotel | **I** India | **J** Juliett | **K** Kilo | **L** Lima | **M** Mike |
| **N** November | **O** Oscar | **P** Papa | **Q** Quebec | **R** Romeo | **S** Sierra | **T** Tango | **U** Uniform | **V** Victor | **W** Whiskey | **X** X-ray | **Y** Yankee | **Z** Zulu |

## HOW TO CHECK INTO A NET

1. Listen first — wait for net control to call for check-ins.
2. Key up, pause 1/2 second (let the repeater open), then talk.
3. Give: call sign (phonetically) · first name · city.
4. Add "no traffic" — or briefly state what you have.
5. UNKEY, then listen — net control will acknowledge you.

## REPEATER ETIQUETTE & TROUBLESHOOTING

- Pause after keying — repeaters take ~1/2 s to come up.
- Leave 2 s between overs so others can break in.
- ID with your call sign every 10 min and at sign-off.
- Can't hear it? Out of range, or it needs a receive tone.

## SKYWARN — severe weather nets

| | | | | |
|---|---|---|---|---|
| **NBL145** | 350 | 145.170 | Hamilton Co (home) | PL77.0 |
| **NBL355** | 633 | 443.550 | Hamilton Co UHF | PL77.0 |
| **IND970** | 346 | 146.970 | Central IN — primary | PL107.2 |
| **IND265** | 621 | 442.650 | Central IN UHF | PL77.0 |
| **IND760** | 344 | 146.760 | Marion Co — primary | PL151.4 |
| **IND325** | 623 | 443.250 | Marion Co — backup | PL100.0 |

## SKYWARN — WHAT TO REPORT

- Tornado, funnel cloud, or wall cloud
- Hail 1 inch (quarter) or larger
- Wind 58 mph+, or any wind damage
- Flash flooding / water over a roadway

**Report: what · where (town + county) · when · your call sign**

Listen first — never break into an active report.

## LINKED — EchoLink / IRLP / AllStar

| | | | | |
|---|---|---|---|---|
| **NBL355** | 633 | 443.550 | Noblesville K9EOC | IRLP · 77.0 |
| **IND970** | 346 | 146.970 | Indianapolis W9ICE | IRLP · 107.2 |
| **IND131** | 340 | 147.315 | Indianapolis K9DC | IRLP · 94.8 |
| **IND342** | 625 | 443.425 | Indianapolis K9IP | IRLP · D043 |
| **IND880** | 345 | 146.880 | Indianapolis W9RCA | EchoLink · 88.5 |
| **IND121** | 339 | 147.210 | Indianapolis W9FBZ | EchoLink · 88.5 |
| **IND332** | 624 | 443.325 | Indianapolis W9EIB | EchoLink · 156.7 |
| **AND775** | 330 | 146.775 | Anderson W2NAP | AllStar+Echo · 88.5 |
| **AND422** | 613 | 442.275 | Anderson W2NAP | AllStar · 151.4 |
| **NCS739** | 353 | 147.390 | New Castle W9OB | AllStar · 127.3 |
| **IND745** | 356 | 146.745 | Indianapolis W9AMT | AllStar 662640 · D346 |

**Courtesy: register w/ DREGs on myGMRS · INDY 600 asks $25/yr donation for regular use.**

Listening is always legal. Ham = RX-only until Technician.

NOAA weather: hold 0 . Scan=Manual → skipped, still selectable.

## US BAND PLAN — quick ranges (MHz)

HAM 160m 1.8–2 · 80m 3.5–4 · 40m 7–7.3 · 20m 14–14.35 · 15m 21–21.45
10m 28–29.7 · 6m 50–54 · 2m 144–148 · 1.25m 222–225 · 70cm 420–450
Tech HF: all of 10m voice 28.3–28.5 + CW slices on 80/40/15
OTHER Air 108–137 AM · MURS 151.8/154.6 · Marine 156–157 · WX 162.4–.55

## MESHCORE — off-grid text mesh (LoRa 902–928 · licence-free)

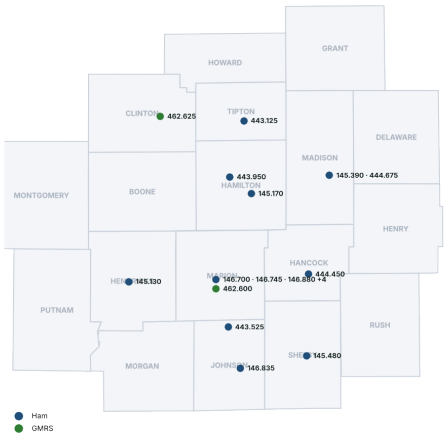
**Indiana preset: 910.525 · BW 62.5 · SF 7 · CR 5 · Hash 2B · TX 22 dBm**

Flash at flasher.meshcore.io — Companion = handheld · Repeater = fixed node

#public · #okimesh · #info · #test · #emergency · #skywarn · #hamradio
#weather · #meshcore · #porchlight — send !echo in #test to check reach

**CIMN cimesh.net (FREE rptr hardware, prime sites) · okimesh.org · map.okimesh.org**

Net repeaters by county, Indianapolis metro



# VHF / UHF Handheld — Channel & Net Reference

RX 108–136 (AM air) · 136–174 · 350–390 · 400–520 MHz  TX 136–174 · 400–480 MHz (FM, 5 W)



● **BOLD / shaded = high-activity: net repeaters · SKYWARN · linked (EchoLink/IRLP/AllStar) · open GMRS repeaters · calling & emergency channels**

| | | | | |
|---|---|---|---|---|
| **NOAA Weather** | | | | |
| ● | 001 | **WXINDY** | **162.5500** | NOAA Wx - 162.550 is Indianapolis p… |
| **Weather / Data Satellites** | | | | |
| | 002 | **ORBCM1** | 137.2500 | ORBCOMM satellite data bursts |
| | 003 | **ORBCM2** | 137.7400 | ORBCOMM satellite data bursts |
| | 004 | **WXSAT1** | 137.1000 | NOAA-19 APT - SDR to decode, hear w… |
| | 005 | **WXSAT2** | 137.6200 | NOAA-15 APT |
| | 006 | **WXSAT3** | 137.9125 | NOAA-18 APT |
| | 007 | **WXSAT4** | 137.9000 | Meteor-M LRPT wx-sat |
| **Satellites (Voice / ISS)** | | | | |
| | 012 | **ISS** | 145.8000 | ISS voice downlink when crew active |
| | 013 | **ISSAPRS** | 145.8250 | ISS/sat APRS digipeater data |
| | 014 | **ISSRPTR** | 437.8000 | ISS crossband FM voice repeater dow… |
| | 015 | **PO101** | 145.9000 | PO-101 Diwata-2 FM (scheduled passe… |
| | 016 | **SO50** | 436.7950 | SO-50 FM satellite downlink |
| | 017 | **TEVEL** | 436.4000 | TEVEL FM repeater cubesats (~436.4) |
| **MURS** | | | | |
| | 020 | **MURS1** | 151.8200 | MURS license-free |
| | 021 | **MURS2** | 151.8800 | MURS license-free |
| | 022 | **MURS3** | 151.9400 | MURS license-free |
| | 023 | **MURS4** | 154.5700 | MURS license-free |
| | 024 | **MURS5** | 154.6000 | MURS license-free |
| **Marine** | | | | |
| | 030 | **MAR09** | 156.4500 | Marine Ch09 hailing/recreational |
| | 031 | **MAR13** | 156.6500 | Marine Ch13 bridge/navigation |
| ● | 032 | **MAR16** | **156.8000** | Marine Ch16 distress/calling |
| | 033 | **MAR22A** | 157.1000 | Marine Ch22A USCG liaison |
| | 034 | **MAR68** | 156.4250 | Marine Ch68 recreational |
| | 035 | **MAR69** | 156.4750 | Marine Ch69 recreational |
| | 036 | **MAR71** | 156.5750 | Marine Ch71 recreational |
| | 037 | **MAR72** | 156.6250 | Marine Ch72 recreational |
| | 038 | **MAR78** | 156.9250 | Marine Ch78 recreational |
| **Railroad** | | | | |
| | 040 | **CSXdsp** | 160.3200 | CSX dispatcher Big Four Yard |
| | 041 | **CSXrd** | 160.2300 | CSX road - Avon/St Louis Line |
| | 042 | **EOT1** | 452.9375 | Railroad EOT/DPU telemetry (data bu… |
| | 043 | **EOT2** | 457.9375 | Railroad EOT/HTU telemetry (data bu… |
| | 044 | **INRDrd** | 160.7700 | Indiana Rail Road road 44-44 |
| | 045 | **INRDyd** | 161.4300 | Indiana Rail Road yard 88-88 |
| | 046 | **NS2** | 160.9500 | Norfolk Southern road AAR42 |
| | 047 | **NS3** | 161.1900 | Norfolk Southern road AAR58 |
| | 048 | **NS4** | 161.2500 | Norfolk Southern road AAR62 |
| | 049 | **NSrd** | 160.8000 | Norfolk Southern road AAR46 |
| **Interop / EMS / Fire** | | | | |
| ● | 028 | **FEDNC1** | **169.5375** | Federal interop NC-1/IR-5 calling -… |
| | 029 | **FEDLE** | 167.2500 | Federal LE interop LE-2/LE-6 (NIFOG) |
| | 050 | **SAR160** | 155.1600 | National Search & Rescue |
| | 051 | **UCALL40** | 453.2125 | UHF national interop calling UCALL40 |
| | 052 | **UTAC41** | 453.4625 | UHF national interop tactical UTAC41 |
| | 053 | **UTAC42** | 453.7125 | UHF national interop tactical UTAC42 |
| | 054 | **UTAC43** | 453.8625 | UHF national interop tactical UTAC43 |
| | 055 | **VCALL10** | 155.7525 | National interop calling VCALL10 |
| | 056 | **VFIR21** | 154.2800 | Fire mutual aid VFIRE21 |
| | 057 | **VFIR22** | 154.2650 | Fire mutual aid VFIRE22 |
| | 058 | **VFIR23** | 154.2950 | Fire mutual aid VFIRE23 |
| | 059 | **VLAW31** | 155.4750 | Law enforcement interop calling |
| | 060 | **VMED28** | 155.3400 | EMS/hospital national HEAR |
| | 061 | **VTAC11** | 151.1375 | National interop VTAC11 (TX PL156.7) |
| | 062 | **VTAC12** | 154.4525 | National interop VTAC12 |
| | 063 | **VTAC13** | 158.7375 | National interop VTAC13 |
| | 064 | **VTAC14** | 159.4725 | National interop VTAC14 |
| | 065 | **VTAC17** | 161.8500 | National interop VTAC17 |
| | 066 | **VFIR24** | 154.2725 | Fire mutual aid VFIRE24 (NIFOG) |
| | 067 | **VFIR25** | 154.2875 | Fire mutual aid VFIRE25 (NIFOG) |
| | 068 | **VFIR26** | 154.3025 | Fire mutual aid VFIRE26 (NIFOG) |
| | 069 | **VMED29** | 155.3475 | EMS interop VMED29 secondary (NIFOG) |
| | 078 | **VLAW32** | 155.4825 | Law enforcement interop VLAW32 seco… |
| | 098 | **SAR123** | 123.1000 | Aeronautical SAR on-scene 123.1 AM … |
| **UHF Hospital (MED)** | | | | |
| | 070 | **MED1** | 463.0000 | UHF EMS/hospital MED-1 |
| | 071 | **MED2** | 463.0250 | UHF EMS/hospital MED-2 |
| | 072 | **MED3** | 463.0500 | UHF EMS/hospital MED-3 |
| | 073 | **MED4** | 463.0750 | UHF EMS/hospital MED-4 |
| | 074 | **MED5** | 463.1000 | UHF EMS/hospital MED-5 |
| | 075 | **MED6** | 463.1250 | UHF EMS/hospital MED-6 |
| | 076 | **MED7** | 463.1500 | UHF EMS/hospital MED-7 |
| | 077 | **MED8** | 463.1750 | UHF EMS/hospital MED-8 |
| **Aircraft** | | | | |
| | 080 | **AIRAIR** | 122.7500 | Air-to-air |
| | 081 | **CAP** | 148.1500 | Civil Air Patrol |
| ● | 082 | **GUARD** | **121.5000** | Aircraft emergency / guard |
| | 083 | **INDapp** | 127.1500 | Indianapolis Approach (east) |
| | 084 | **KEYEct** | 122.8000 | Eagle Creek CTAF/UNICOM |
| | 085 | **KINDap** | 128.7500 | Indianapolis Intl Approach |
| | 086 | **KINDat** | 134.2500 | Indianapolis Intl ATIS |
| | 087 | **KINDdp** | 119.0500 | Indianapolis Intl Departure |
| | 088 | **KINDgd** | 121.8000 | Indianapolis Intl Ground |
| | 089 | **KINDtw** | 120.9000 | Indianapolis Intl Tower |
| | 090 | **KMQJcl** | 119.2500 | Mt Comfort Clearance |
| | 091 | **KMQJct** | 122.7000 | Mt Comfort CTAF |
| | 092 | **KMQJwx** | 124.1750 | Mt Comfort AWOS |
| | 093 | **KUMPcl** | 121.6250 | Indy Metro Clearance |
| | 094 | **KUMPct** | 123.0000 | Indy Metro/Fishers CTAF |
| | 095 | **KUMPwx** | 119.3750 | Indy Metro/Fishers AWOS |
| | 096 | **MULTI** | 122.9000 | Multicom (uncontrolled fields) |
| | 097 | **UNICOM** | 122.9500 | UNICOM (towered GA) |
| **FRS / GMRS** | | | | |
| | 100 | **G01-562** | 462.5625 | FRS/GMRS ch1 - HCEM community emerg.. |
| | 101 | **G02-587** | 462.5875 | FRS/GMRS ch2 |
| | 102 | **G03-612** | 462.6125 | FRS/GMRS ch3 |
| | 103 | **G04-637** | 462.6375 | FRS/GMRS ch4 |
| | 104 | **G05-662** | 462.6625 | FRS/GMRS ch5 |
| | 105 | **G06-687** | 462.6875 | FRS/GMRS ch6 |
| | 106 | **G07-712** | 462.7125 | FRS/GMRS ch7 |
| | 107 | **F08-562** | 467.5625 | FRS/GMRS ch8 |
| | 108 | **F09-587** | 467.5875 | FRS/GMRS ch9 |
| | 109 | **F10-612** | 467.6125 | FRS/GMRS ch10 |
| | 110 | **F11-637** | 467.6375 | FRS/GMRS ch11 |
| | 111 | **F12-662** | 467.6625 | FRS/GMRS ch12 |
| | 112 | **F13-687** | 467.6875 | FRS/GMRS ch13 |
| | 113 | **F14-712** | 467.7125 | FRS/GMRS ch14 |
| | 114 | **G15-550** | 462.5500 | FRS/GMRS ch15 (GMRS repeater output) |
| ● | 115 | **ICE550** | **462.5500** | 77.0 ICE 550 GMRS rptr WQPV948 (Indianap… |
| | 116 | **I15-550** | 467.5500 | GMRS repeater INPUT for ch15 - list… |
| | 117 | **G16-575** | 462.5750 | FRS/GMRS ch16 (GMRS repeater output) |
| | 118 | **BURL575** | 462.5750 | 131.8 Burlington GMRS rptr WQSK928 (Carro… |
| | 119 | **ELWD575** | 462.5750 | 136.5 Elwood GMRS rptr WROI852 (Madison) … |
| ● | 120 | **DREG575** | **462.5750** | D043 DREGs 575 (Mighty 575) GMRS rptr ch… |
| | 121 | **I16-575** | 467.5750 | GMRS repeater INPUT for ch16 - list… |
| ● | 122 | **G17-600** | **462.6000** | FRS/GMRS ch17 (GMRS repeater output) |
| | 123 | **I17-600** | 467.6000 | GMRS repeater INPUT for ch17 - list… |
| | 124 | **G18-625** | 462.6250 | FRS/GMRS ch18 (GMRS repeater output) |
| | 125 | **ALEX625** | 462.6250 | 123.0 Alexandria GMRS rptr WRKY635 (Madis… |
| ● | 126 | **MTWN625** | **462.6250** | 131.8 Michigantown 625 GMRS rptr WROU622 … |
| | 127 | **I18-625** | 467.6250 | GMRS repeater INPUT for ch18 - list… |
| ● | 128 | **G19-650** | **462.6500** | FRS/GMRS ch19 (GMRS repeater output) |
| ● | 129 | **MOS650** | **462.6500** | 141.3 Moses GMRS rptr WSFW943 (Lizton, He… |
| | 130 | **WLF650** | 462.6500 | D664 TRAVEL - West Lafayette 650 GMRS rp… |
| ● | 131 | **CAL-650** | **462.6500** | 141.3 GMRS ch19 road-travel simplex - 141… |
| | 132 | **I19-650** | 467.6500 | GMRS repeater INPUT for ch19 - list… |
| | 133 | **G20-675** | 462.6750 | FRS/GMRS ch20 (GMRS repeater output) |
| ● | 134 | **ANDR675** | **462.6750** | 167.9 Anderson GMRS rptr WSFQ575 MCEMA (M… |
| ● | 135 | **AVON675** | **462.6750** | 136.5 Avon GMRS rptr WSHP321 (Hendricks) … |
| | 136 | **FLSH675** | 462.6750 | PLACEHOLDER Flashes 675 GMRS rptr (… |
| | 137 | **I20-675** | 467.6750 | GMRS repeater INPUT for ch20 - list… |
| | 138 | **G21-700** | 462.7000 | FRS/GMRS ch21 (GMRS repeater output) |
| | 139 | **RUSS700** | 462.7000 | 131.8 Russiaville GMRS rptr WQSK928 (Howa… |
| | 140 | **BRWN700** | 462.7000 | D411 TRAVEL - Brown County 700 GMRS rptr… |
| | 141 | **BBG700** | 462.7000 | PLACEHOLDER 502 Repeater GMRS rptr … |
| | 142 | **AVN700** | 462.7000 | PLACEHOLDER Old School Indy GMRS rp… |
| | 143 | **HEN700** | 462.7000 | 151.4 Hendricks County GMRS rptr ch21 - A… |
| | 144 | **I21-700** | 467.7000 | GMRS repeater INPUT for ch21 - list… |
| | 145 | **G22-725** | 462.7250 | FRS/GMRS ch22 (GMRS repeater output) |
| ● | 146 | **SHAK725** | **462.7250** | 118.8 Shackleford725 K931 GMRS rptr (NE I… |
| | 147 | **I22-725** | 467.7250 | GMRS repeater INPUT for ch22 - list… |
| ● | 148 | **CAL-675** | **462.6750** | 141.3 GMRS ch20 road-travel/emergency cal… |
| ● | 999 | **LNGHAUL** | **462.6750** | 141.3 Project LongHaul calling - 462.675 … |
| **Business / Itinerant** | | | | |
| | 200 | **BIZ01** | 151.6250 | Itinerant red dot - retail/business |
| | 201 | **BIZ02** | 151.7000 | Business itinerant |
| | 202 | **BIZ03** | 151.7600 | Business itinerant |
| | 203 | **BIZ04** | 151.9550 | Itinerant purple dot / also Noblesv… |
| | 204 | **BIZ05** | 158.4000 | Itinerant white dot |
| | 205 | **BIZ06** | 464.5000 | Itinerant brown dot - stores |
| | 206 | **BIZ07** | 464.5500 | Itinerant yellow dot |
| | 207 | **BIZ08** | 469.5000 | Itinerant brown dot input |
| | 208 | **BIZ09** | 469.5500 | Itinerant yellow dot input |
| | 209 | **BIZ10** | 467.8500 | UHF itinerant |
| | 210 | **BIZ11** | 467.8750 | UHF itinerant |

# VHF / UHF Handheld — Channel & Net Reference

RX 108–136 (AM air) · 136–174 · 350–390 · 400–520 MHz  TX 136–174 · 400–480 MHz (FM, 5 W)



● **BOLD / shaded = high-activity: net repeaters · SKYWARN · linked (EchoLink/IRLP/AllStar) · open GMRS repeaters · calling & emergency channels**

| | | | | |
|---|---|---|---|---|
| 211 | **BIZ12** | 467.9000 | — | UHF itinerant |
| 212 | **BIZ13** | 467.9250 | — | UHF itinerant |
| 213 | **BIZ14** | 451.8000 | — | UHF business |
| 214 | **BIZ15** | 452.9000 | — | UHF business |
| 215 | **BIZ16** | 461.0000 | — | UHF business |
| 216 | **BIZ17** | 464.4750 | — | UHF business |
| 217 | **BIZ18** | 464.5250 | — | UHF business |
| **Venues / Events** | | | | |
| 220 | **VEN01** | 151.5125 | — | Venue/retail/event itinerant - scan… |
| 221 | **VEN02** | 151.6325 | — | Venue/retail/event itinerant - scan… |
| 222 | **VEN03** | 154.5275 | — | Venue/retail/event itinerant - scan… |
| 223 | **VEN04** | 457.5375 | — | Venue/retail/event itinerant - scan… |
| 224 | **VEN05** | 457.5625 | — | Venue/retail/event itinerant - scan… |
| 225 | **VEN06** | 457.6125 | — | Venue/retail/event itinerant - scan… |
| 226 | **VEN07** | 461.0375 | — | Venue/retail/event itinerant - scan… |
| 227 | **VEN08** | 461.0625 | — | Venue/retail/event itinerant - scan… |
| 228 | **VEN09** | 467.7625 | — | Venue/retail/event itinerant - scan… |
| 229 | **VEN10** | 467.8125 | — | Venue/retail/event itinerant - scan… |
| 230 | **VEN11** | 469.2625 | — | Venue/retail/event itinerant - scan… |
| **Speedway (IMS)** | | | | |
| 240 | **IMS1** | 454.0000 | — | IMS race team UHF - scan during Spe… |
| 241 | **IMS2** | 456.5000 | — | IMS race team UHF - scan during Spe… |
| 242 | **IMS3** | 458.0000 | — | IMS race team UHF - scan during Spe… |
| 243 | **IMS4** | 461.5000 | — | IMS race team UHF - scan during Spe… |
| **Public Svc / Schools** | | | | |
| 245 | **HSE461** | 461.3250 | — | HSE Schools ops 1 WPDU924 (Fishers) |
| 246 | **HSE462** | 461.8500 | — | HSE Schools ops 2 WPDU924 (Fishers) |
| 247 | **HSE463** | 462.3250 | — | HSE Schools ops 3 WPDU924 (Fishers)… |
| 248 | **HSEBUS** | 463.3250 | — | HSE School BUSES WPDU924 (Fishers) … |
| 249 | **HSE464** | 463.4000 | — | HSE Schools ops 4 WPDU924 (Fishers)… |
| 250 | **NOBBUS** | 155.1750 | — | Noblesville School buses WQDY357 |
| 251 | **NOBSC1** | 152.9450 | — | Noblesville Schools ops WQHJ807 |
| 252 | **NOBCRS** | 154.5400 | — | Noble Crossing Elementary WQHH509 |
| 253 | **STNYCK** | 152.3600 | — | Stony Creek Elementary WQBJ508 |
| 254 | **CARGOV** | 453.2375 | — | Carmel Government Services WPXH695 |
| 255 | **CARPRK** | 453.1500 | — | Carmel Parks / Monon Center WQIA462 |
| 256 | **CARUTA** | 155.8650 | — | Carmel Utility A KSZ831 |
| 257 | **CARUTB** | 453.7500 | — | Carmel Utility B KNIM261 |
| 258 | **CARUTC** | 453.8750 | — | Carmel Utility C WNIF564 |
| 259 | **OLMC1** | 457.4125 | — | Our Lady Mt Carmel ops1 WQOX476 PL1… |
| 260 | **OLMC2** | 457.8875 | — | Our Lady Mt Carmel ops2 WQOX476 PL1… |
| 261 | **HHSCV** | 151.7150 | — | Hamilton Heights Schools VHF WPWF63… |
| 262 | **HHSCU** | 462.2000 | — | Hamilton Heights Schools UHF WPWF63… |
| 263 | **SIRHAM** | 460.5250 | — | Hamilton Co outdoor warning siren D… |
| 264 | **SIRCAR** | 453.6000 | — | Carmel weather siren DATA WPML424 (… |
| **Ham 2 m** | | | | |
| 300 | **144200** | 144.2000 | — | 2m SSB DX/weak-signal calling - SSB… |
| 301 | **145500** | 145.5000 | — | IARU R1 2m FM calling (internationa… |
| 302 | **146400** | 146.4000 | — | 2m FM simplex |
| 303 | **146415** | 146.4150 | — | 2m FM simplex |
| 304 | **146430** | 146.4300 | — | 2m FM simplex |
| 305 | **146445** | 146.4450 | — | 2m FM simplex |
| 306 | **146475** | 146.4750 | — | 2m FM simplex |
| 307 | **146490** | 146.4900 | — | 2m FM simplex |
| 308 | **146505** | 146.5050 | — | 2m FM simplex |
| 309 | **146535** | 146.5350 | — | 2m FM simplex |
| 310 | **146565** | 146.5650 | — | 2m FM simplex |
| 311 | **147435** | 147.4350 | — | 2m FM simplex |
| 312 | **147450** | 147.4500 | — | 2m FM simplex |
| 313 | **147465** | 147.4650 | — | 2m FM simplex |
| 314 | **147495** | 147.4950 | — | 2m FM simplex |
| 315 | **147525** | 147.5250 | — | 2m FM simplex |
| 316 | **147570** | 147.5700 | — | 2m FM simplex |
| 317 | **147585** | 147.5850 | — | 2m FM simplex |
| 318 | **2M-46** | 146.4600 | — | 2m simplex |
| 319 | **2M-55** | 146.5500 | — | 2m simplex |
| 320 | **2M-58** | 146.5800 | — | 2m simplex |
| 321 | **2M-742** | 147.4200 | — | 2m simplex |
| 322 | **2M-748** | 147.4800 | — | 2m simplex |
| 323 | **2M-751** | 147.5100 | — | 2m simplex |
| 324 | **2M-754** | 147.5400 | — | 2m simplex |
| 325 | **2M-755** | 147.5550 | — | 2m simplex |
| ● 326 | **2MCALL** | **146.5200** | — | 2m national calling |
| 327 | **AND090** | 147.0900 | 110.9 | Anderson K9DYR IRLP (Madison) PL110… |
| 328 | **AND150** | 147.1500 | 151.4 | Anderson KD9SZA (Madison) PL151.4 |
| ● 329 | **AND390** | **145.3900** | 151.4 | Anderson Hospital W9VCF (Madison) P… |
| ● 330 | **AND775** | **146.7750** | 88.5 | Anderson W2NAP AllStar/EchoLink (Ma… |
| 331 | **AND820** | 146.8200 | 110.9 | Anderson Madison Co Dispatch W9OBH … |
| 332 | **APRS** | 144.3900 | — | APRS data (national) - digital burs… |
| 333 | **BEL147** | 147.0150 | 88.5 | Belleville N9HC (Hendricks) PL88.5 |
| ● 334 | **DAN145** | **145.1300** | 88.5 | Danville W9HCA (Hendricks) PL88.5 |
| ● 335 | **FRA146** | **146.8350** | 151.4 | Franklin (Johnson Co) repeater - Su… |
| 336 | **GFD145** | 145.3300 | 88.5 | Greenfield W9ATG (Hancock) PL88.5 |
| 337 | **HOR146** | 146.8650 | 77.0 | Hortonville K9EOC (Hamilton) PL77.0 |
| 338 | **IND112** | 147.1200 | 88.5 | Indianapolis W9IRA PL88.5 |
| ● 339 | **IND121** | **147.2100** | 88.5 | Indianapolis W9FBZ Echo PL88.5 |
| ● 340 | **IND131** | **147.3150** | 94.8 | Indianapolis K9DC IRLP PL94.8 |
| 341 | **IND625** | 146.6250 | 88.5 | Indianapolis W9IRA Fusion PL88.5 |
| 342 | **IND670** | 146.6700 | 88.5 | Indianapolis K9IPL Fusion PL88.5 |
| ● 343 | **IND700** | **146.7000** | 88.5 | Indianapolis W9IRA WTHR tower PL88.5 |
| ● 344 | **IND760** | **146.7600** | 151.4 | Indianapolis K9LPW Fusion PL151.4 \|… |
| ● 345 | **IND880** | **146.8800** | 88.5 | Indianapolis W9RCA Echo/Fusion PL88… |
| ● 346 | **IND970** | **146.9700** | 107.2 | Indianapolis W9ICE IRLP PL107.2 \| S… |
| 347 | **KEM145** | 145.2800 | 131.8 | Kempton K9TRC (Tipton) PL131.8 |
| 348 | **LEB147** | 147.1050 | 77.0 | Lebanon N9HLL (Boone) PL77.0 |
| 349 | **MTC147** | 147.1650 | 88.5 | Montclair KO9F (Hendricks) PL88.5 |
| ● 350 | **NBL145** | **145.1700** | 77.0 | Noblesville K9EOC (Hamilton) PL77.0… |
| 351 | **NCS145** | 145.4500 | 127.3 | New Castle NT9J (Henry) PL127.3 |
| ● 353 | **NCS739** | **147.3900** | 127.3 | New Castle W9OB AllStar (Henry) PL1… |
| 354 | **STL145** | 145.2900 | 88.5 | Stilesville AB9D (Hendricks) PL88.5 |
| 355 | **SUM310** | 145.3100 | 131.8 | Summitville WB9UCF (Madison) PL131.8 |
| ● 356 | **IND745** | **146.7450** | D346 | Indianapolis W9AMT AllStar (Marion)… |
| 357 | **MRV110** | 145.1100 | 88.5 | Mooresville W9WWV (Morgan) PL88.5 |
| ● 358 | **SHB480** | **145.4800** | 88.5 | Shelbyville W9NTP (Shelby) PL88.5 |
| 360 | **MTV706** | 147.0600 | 88.5 | Martinsville K9PYI (Morgan) CSQ |
| 361 | **MTV725** | 147.2550 | 88.5 | Martinsville W9ZSK (Morgan) PL88.5 |
| 362 | **MGN230** | 145.2300 | 151.4 | Morgan Co K4IO (Martinsville area) … |
| 363 | **RSH147** | 147.0000 | 127.3 | Rushville W9NTP (Rush Co ~40mi) PL1… |
| 364 | **NBLDIG** | 145.0900 | — | Hamilton Co EM packet/APRS digipeat… |
| **Ham 70 cm** | | | | |
| 218 | **CPR464** | 464.1125 | — | Conner Prairie 1 ops WQLH307 (Fishe… |
| 219 | **CPR463** | 463.6375 | — | Conner Prairie 3 ops WQLH307 (Fishe… |
| 231 | **CPRdmr** | 463.7250 | — | Conner Prairie 2 ops WQLH307 (Fishe… |
| 600 | **432100** | 432.1000 | — | 70cm SSB DX/weak-signal calling - S… |
| 601 | **433500** | 433.5000 | — | 70cm FM calling (international) / c… |
| 602 | **445950** | 445.9500 | — | 70cm FM simplex |
| 603 | **445975** | 445.9750 | — | 70cm FM simplex |
| 604 | **446125** | 446.1250 | — | 70cm FM simplex |
| 605 | **446175** | 446.1750 | — | 70cm FM simplex |
| 606 | **70-025** | 446.0250 | — | 70cm simplex |
| 607 | **70-050** | 446.0500 | — | 70cm simplex |
| 608 | **70-075** | 446.0750 | — | 70cm simplex |
| 609 | **70-100** | 446.1000 | — | 70cm simplex |
| 610 | **70-150** | 446.1500 | — | 70cm simplex |
| 611 | **70-200** | 446.2000 | — | 70cm simplex |
| ● 612 | **70CALL** | **446.0000** | — | 70cm national calling |
| ● 613 | **AND422** | **442.2750** | 151.4 | Anderson W2NAP AllStar (Madison) PL… |
| 614 | **AND433** | 443.3500 | 110.9 | Anderson Fire Trng WA9CWE (Madison)… |
| 615 | **AND446** | 444.6750 | — | Anderson WA9EOC (Madison) CSQ |
| 616 | **ARC362** | 443.6250 | 88.5 | Arcadia K3HTK Fusion (Hamilton) PL8… |
| ● 617 | **GFD444** | **444.4500** | D465 | Greenfield W9ATG (Hancock) DCS465 |
| 618 | **GWD423** | 442.3500 | 151.4 | Greenwood KQ9Y Fusion (Johnson) PL1… |
| 619 | **GWD435** | 443.5250 | 151.4 | Greenwood W9MID Fusion (Johnson) PL… |
| ● 620 | **HOR395** | **443.9500** | 77.0 | Hortonville K9EOC WIRES-X (Hamilton… |
| ● 621 | **IND265** | **442.6500** | 77.0 | Indianapolis KM9E PL77.0 \| SKYWARN … |
| 622 | **IND300** | 443.0000 | 100.0 | Indianapolis WB9PGW PL100.0 |
| ● 623 | **IND325** | **443.2500** | 100.0 | Indianapolis KM9E PL100.0 \| SKYWARN… |
| ● 624 | **IND332** | **443.3250** | 156.7 | Indianapolis W9EIB Echo PL156.7 |
| ● 625 | **IND342** | **443.4250** | D043 | Indianapolis K9IP IRLP DCS043 |
| 626 | **IND375** | 443.7500 | 100.0 | Indianapolis K9LPW WIRES-X PL100.0 |
| 627 | **IND380** | 443.8000 | — | Indianapolis KC9COP CSQ |
| 628 | **IND385** | 443.8500 | 88.5 | Indianapolis W9IRA WTHR PL88.5 |
| 629 | **IND400** | 444.0000 | 100.0 | Indianapolis NE9T PL100.0 |
| 630 | **IND432** | 444.3250 | 136.5 | Indianapolis KB9SGN PL136.5 |
| 632 | **MTC442** | 442.9000 | 88.5 | Montclair KO9F Fusion (Hendricks) P… |
| ● 633 | **NBL355** | **443.5500** | 77.0 | Noblesville K9EOC (Hamilton) PL77.0… |
| 634 | **NCS441** | 441.4000 | 141.3 | New Castle K9TDX (Henry) PL141.3 |
| 635 | **NCS444** | 444.2750 | 127.3 | New Castle NT9J (Henry) PL127.3 |
| 636 | **SHP443** | 443.7000 | 131.8 | Sharpsville K9TRC (Tipton) PL131.8 |
| ● 637 | **TIP443** | **443.1250** | 131.8 | Tipton K9TRC (Tipton) PL131.8 |
| 638 | **ICE390** | 443.9000 | 77.0 | W9ICE 900MHz-system HUB (WB9OPR QTH… |
| 639 | **IND237** | 442.3750 | — | Indianapolis K9DC IRLP (Marion) CSQ |
| 640 | **GFD472** | 444.7250 | — | Greenfield W9ATG (Hancock) 444.725 … |
| 641 | **EDB025** | 443.0250 | 103.5 | Edinburgh N9JWR (Shelby) PL103.5 |
| 642 | **MTV425** | 444.2500 | 100.0 | Martinsville W9ZSK (Morgan) CSQ |
| 643 | **MTV495** | 444.9500 | 100.0 | Martinsville K9PYI (Morgan) CSQ |
| 644 | **SHB975** | 444.9750 | — | Shelbyville W9NTP (Shelby) CSQ |
| 645 | **NBL25** | 444.6250 | — | NF9K P25 NAC293 (Noblesville) - NEW… |

# VHF / UHF Handheld — Channel & Net Reference

RX 108–136 (AM air) · 136–174 · 350–390 · 400–520 MHz  TX 136–174 · 400–480 MHz (FM, 5 W)

Indianapolis Metro

GMRS ch → GMRS licence · Ham 2m/70cm → Technician · MURS/FRS licence-free



## US AMATEUR BANDS & PRIVILEGES

| Band | MHz | Who can use it |
|---|---|---|
| **160 m** | 1.8 - 2.0 | General+ |
| **80 m** | 3.5 - 4.0 | Gen+ (Tech: CW slice) |
| **60 m** | 5 chans ~5.3 | General+ (shared federal) |
| **40 m** | 7.0 - 7.3 | Gen+ (Tech: CW slice) |
| **30 m** | 10.1 - 10.15 | General+ - CW/digital only |
| **20 m** | 14.0 - 14.35 | General+ |
| **17 m** | 18.068 - 18.168 | General+ |
| **15 m** | 21.0 - 21.45 | Gen+ (Tech: CW slice) |
| **12 m** | 24.89 - 24.99 | General+ |
| **10 m** | 28.0 - 29.7 | **TECH: 28.3-28.5 SSB + CW** |
| **6 m** | 50 - 54 | **Technician+** |
| **2 m** | 144 - 148 | **Technician+** |
| **1.25 m** | 222 - 225 | **Technician+** |
| **70 cm** | 420 - 450 | **Technician+** |
| **33 cm** | 902 - 928 | Tech+ (also MeshCore LoRa) |
| **23 cm** | 1240 - 1300 | **Technician+** |

## LICENSE LADDER

**TECHNICIAN - 35 Q: all VHF/UHF + 10 m voice slice.**
**GENERAL - 35 Q: most of every HF band.**
**AMATEUR EXTRA - 50 Q: everything.**
Study free at hamstudy.org - local exam sessions:
indianahamradio.com/exams - no Morse code required.

## CALLING & ACTIVITY FREQUENCIES

### FM SIMPLEX CALLING

| | |
|---|---|
| 6 m | **52.525** |
| 2 m | **146.520** |
| 1.25 m | **223.500** |
| 70 cm | **446.000** |

### SSB CALLING

| | |
|---|---|
| 10 m (Tech!) | **28.400** |
| 6 m | **50.125** |
| 2 m | **144.200** |
| 70 cm | **432.100** |

### FT8 - busiest mode in ham radio

| | |
|---|---|
| 80/40/30 m | **3.573 / 7.074 / 10.136** |
| 20/17/15 m | **14.074 / 18.100 / 21.074** |
| 12/10/6 m | **24.915 / 28.074 / 50.313** |

### CW / QRP

| | |
|---|---|
| CW QRP | **7.030 / 14.060** |
| SSB QRP | **14.285** |

### DX WATERING HOLE

| | |
|---|---|
| 20 m SSB DX | **14.150 - 14.300** |

## TIME SIGNALS & BEACONS

WWV (AM): 2.5 / 5 / 10 / 15 / 20 MHz - atomic time
CHU Canada: 3.330 / 7.850 / 14.670 MHz
NCDXF beacons: 14.100 / 18.110 / 21.150 / 24.930 / 28.200
Hear a beacon = that band is open to that region.

## Q-CODES - the shorthand you'll hear

| | |
|---|---|
| **QSO** | a contact / conversation |
| **QRZ?** | who is calling me? |
| **QTH** | my location |
| **QSL** | received & confirmed |
| **QRM** | man-made interference |
| **QRN** | static / noise |
| **QRP** | low power operating |
| **QSY** | changing frequency |
| **QRT** | shutting down |
| **QST** | bulletin to all hams |
| **73** | best regards |
| **88** | hugs & kisses |

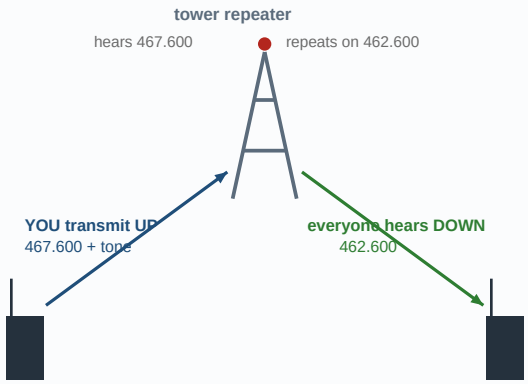
Signal report RST: "59" = readability 5, strength 9

## RESOURCES

| | |
|---|---|
| **arrl.org** | national assoc + band charts |
| **hamstudy.org** | free exam prep |
| **indianahamradio.com** | clubs / nets / exams statewide |
| **repeaterbook.com** | ham repeater directory |
| **mygmrs.com** | GMRS repeaters + network |
| **pota.app** | Parks on the Air spots |
| **brandmeister.network** | DMR + Hoseline listening |
| **netlogger.org** | watch net check-ins live |
| **pigletradio.org** | local club N9BCN, POTA, Discord |
| **HAMCOHAM group** | net reports: groups.google.com/g/hamcoham |
| **okimesh.org** | MeshCore LoRa mesh (region) |

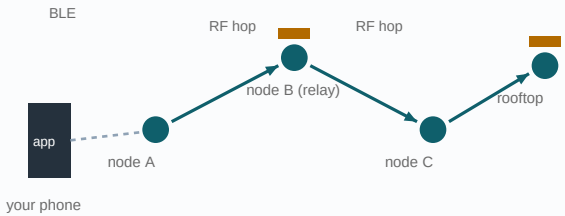
New here? Listen first - then: HamStudy -> Technician exam ->
check into the Who Knows What Net (Thu 8 PM, 443.950 +5, PL 77.0).

## HOW A REPEATER WORKS




offset: UHF +5 MHz · 2 m ±0.6 MHz
tone (PL/DCS) = the key that opens the door
height beats power — a 5 W HT works 40+ mi via tower
pause 1/2 s after keying before you talk

## HOW MESHCORE WORKS (LoRa mesh)




**SETTINGS — flash at flasher.meshcore.io (Companion=handheld, Repeater=fixed)**
**RADIO (Indiana preset): 910.525 · BW 62.5 · SF 7 · CR 5 · Hash 2B · TX 22 dBm**
CHANNELS: #public (default) · #okimesh · #info · test nodes in #test
also: #emergency · #skywarn · #hamradio · #weather
Encrypted text · no internet/towers/licence (902–928 ISM Part 15).
**LOCAL: CIMN (cimesh.net) — FREE repeater hardware for prime sites!**
okimesh.org · cimesh.net · live map: map.okimesh.org · IN Mesh Discord

## DMR IN 4 STEPS (digital voice)

1. Get your ham licence (Technician is enough).
2. Get a free DMR ID at radioid.net — it's your digital call sign.
3. Build the codeplug. Each channel needs FOUR things:
   Frequency · Color Code (like a PL tone) ·
   Time Slot (1 or 2) · Talkgroup (the "room")
4. Key up — the talkgroup routes you: local repeater or a
   $100 hotspot → BrandMeister → the world.

**WHY IT'S DIFFERENT: 2 time slots = 2 conversations on one**
repeater at once. Talkgroups = chat rooms spanning networks
(TG 3118 = all Indiana · TG 31189 = Crossroads · TG 91 = world).
One radio + one hotspot reaches every DMR net on earth.

**LISTEN FREE FIRST: hose.brandmeister.network (no radio needed)**
**DECODE LOCALLY: RTL-SDR + SDRTrunk hears DMR off the air.**
**Local DMR RF: 441.200 Indy · W9AMT 146.745 (mixed FM/DMR).**